1. 35A1 H = 0011 0101 1010 0001
   a) Entry = **6B4 H**, Table = **GDT**, RPL = **01**
   b) Start = **B61310BD**
      G = 0
      End = Start + Limit = B61310BD + 50AEA = **B6181BA7**
   c) Phy Addr = Start + Offset = B61310BD + 158C0 = **B614697D**
   d) Access = 53H = 0101 0011
      A = 1: **accessed**
   e) E = 0, ED = 0 (expand up): **DS**
   f) P = 0: **undefined**

2.
   a)
      i) $\frac{8300608}{2^{24}-1} \times 5V = 2.47V$
      ii) $\frac{1760}{2^{12}-1} \times 100\% = 42.97\%$
   b) SPI: 3-wire system, full-duplex
      I2C: 2-wire system, half-duplex

      <u>Sequence Diagram</u>

      Master ---------- Slave
      ---------- Start ---------->
      ---------- 0000111 + 1 ---------->
      <---------- ACK ----------

      <---------- 01110101 (u)----------
      ---------- ACK ---------->

      <---------- 01000111 (G)----------
      ---------- ACK ---------->

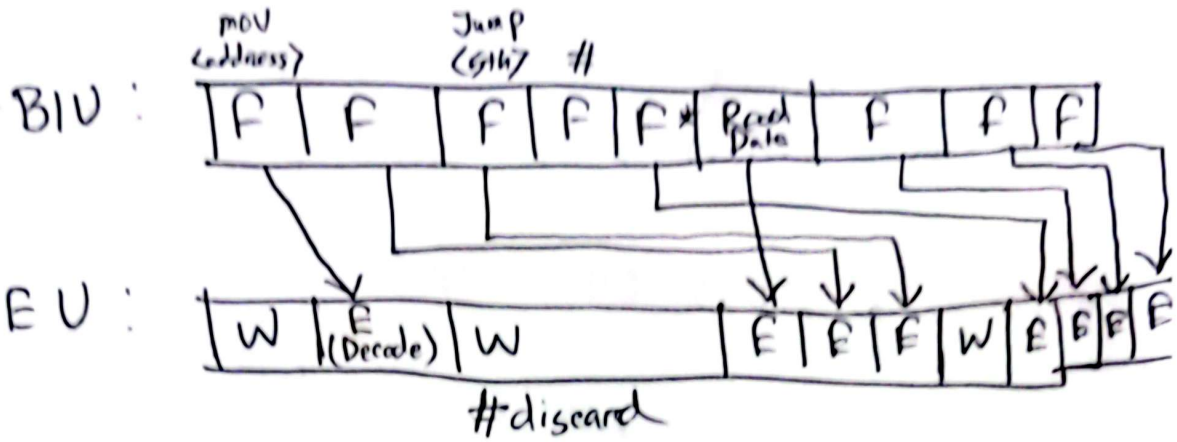
      <---------- 01110010 (r)----------
      ---------- ACK ---------->

      <---------- 01100001 (a)----------
      ---------- NACK ---------->

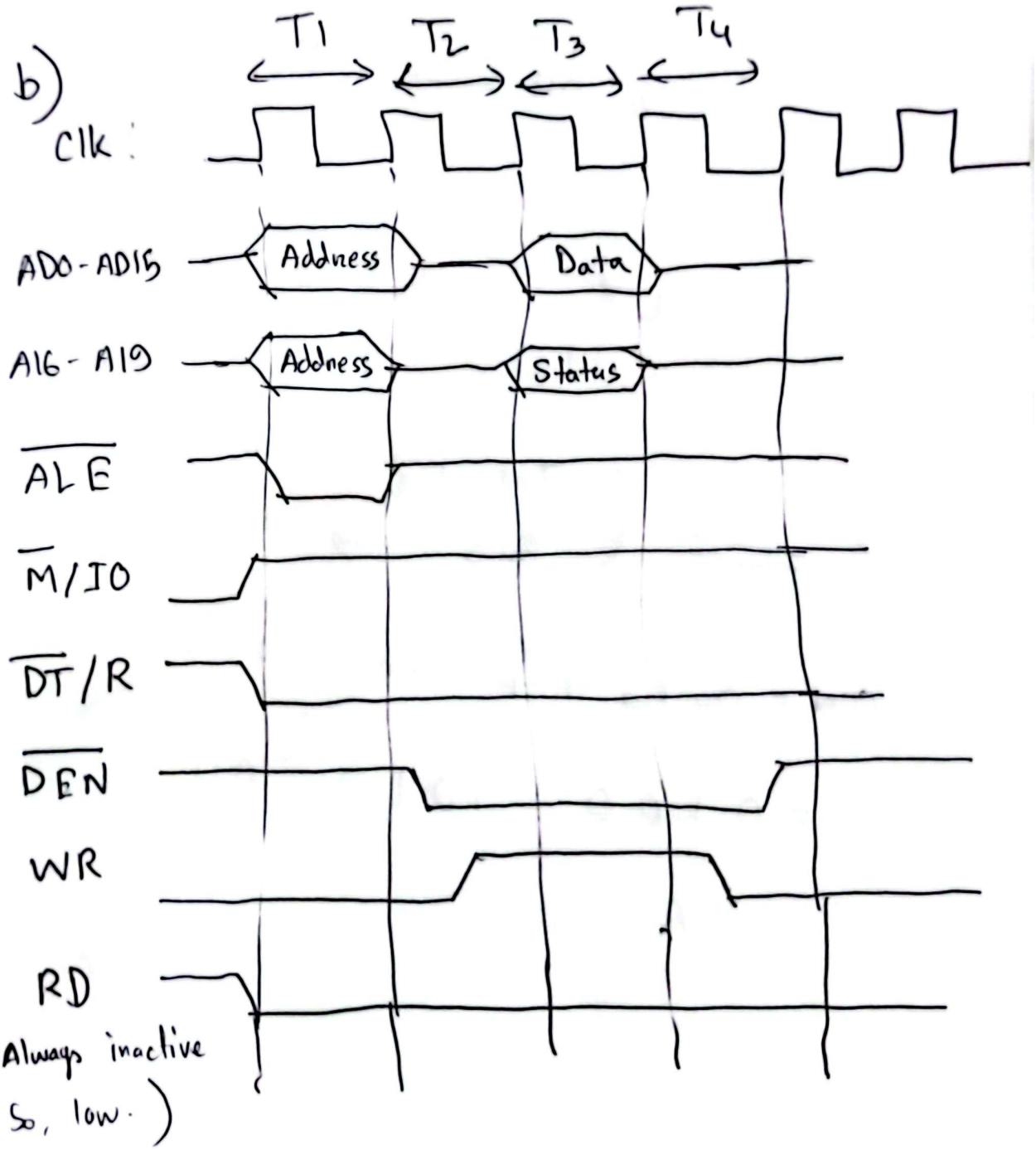
      ---------- Stop ---------->

c) **Raspberry Pi** for dynamic monitoring & adjustments.

3. a)


mov
<address>
Jump
<disp>
#
BIU :
F
Read Data
EU :
W
E (Decode)
#discard

b)


T1
T2
T3
T4
Clk :
AD0 - AD15
Address
Data
A16 - A19
Address
Status
ALE
M/IO
DT/R
DEN
WR
RD
(Always inactive
So, low.)

c) Efficient memory use.

4. a)

	b) AX	BX	Out
Initially ⟶	1H	3H	4H
ADD AH ⟶	3H	2H	4H
ADD CH ⟶	8H	5H	4H
SUB AH ⟶	6H	2H	4H
SUB DH ⟶	5H	1H	4H
OUT ⟶	5H	1H	~~5H~~
HLT ⟶	0H	0H	0H

c) T4 (Ins. Reg → Bus
Bus → MAR)

$CON = S_U \overline{E_U} C_P E_P \quad L_M C_E \overline{L_I} \overline{E_I} \quad L_A \overline{E_A} \overline{L_B} \overline{L_O}$

= 0100 1010 0111

= 4A7H

T5 (RAM → Bus
Bus → B Reg.)

CON = 0100 0111 0101

= 475H

T6 (Add
Adder/Sub → Bus
Bus → AX)

CON = 0000 0011 1111

= 03FH